

1. In Cisco Packet Tracer, set up a simple network with one DHCP server and two client PCs. Configure the DHCP server to assign IP addresses automatically, then verify on each PC that it receives a different IP address from the DHCP pool.

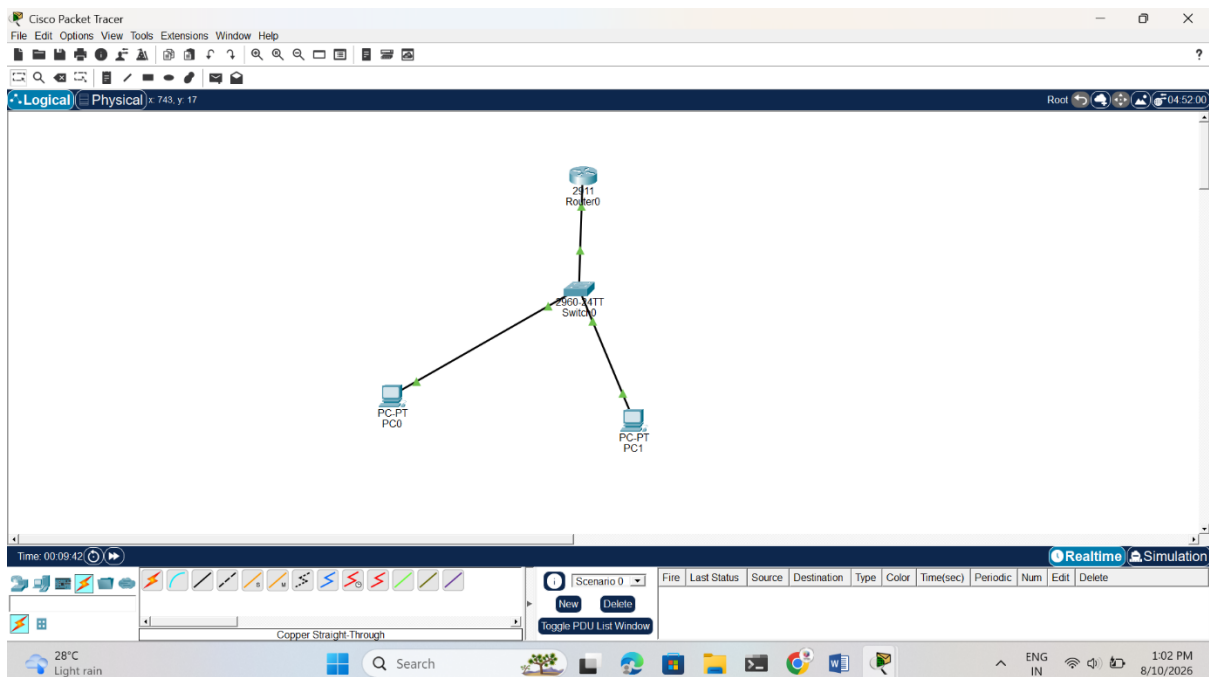
```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip dhcp excluded-address 192.168.1.1 192.168.1.9
^
% Invalid input detected at '^' marker.

Router(config)#ip dhcp excluded-address 192.168.1.1 192.168.1.9
Router(config)#ip dhcp pool LAN
Router(dhcp-config)#network 192.168.1.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.1.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

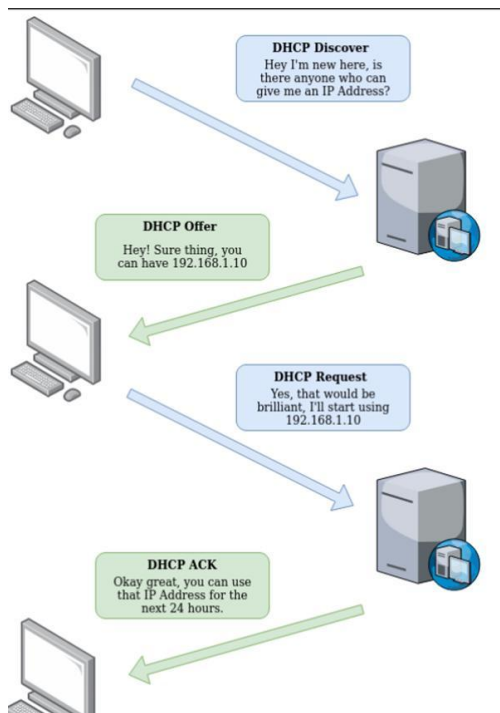
Router#write memory
Router#en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
```



2. Draw a sequence diagram (hand-drawn or digital) that clearly shows each step of the DHCP process: Discover, Offer, Request, and Acknowledge, labeling the sender and receiver at each stage. Hint: Use arrows to show the direction of each message between client and server.



3. Change the DHCP pool range on your Cisco Packet Tracer DHCP server so that only one IP address is available. Add two PCs and observe what happens when both try to get an IP address. Take a screenshot of the result and explain in 2-3 lines why only one PC succeeds.

```

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shut down

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#ip dhcp excluded-address 192.168.1.1 192.168.1.9
Router(config)#ip dhcp pool LAN
Router(dhcp-config)#network 192.168.1.1 255.255.255.0
Router(dhcp-config)#exit
Router(config)#ip dhcp excluded-address 192.168.1.1 192.168.1.254
Router(config)#ip dhcp pool LAN
Router(dhcp-config)#network 192.168.1.1 255.255.255.0
Router(dhcp-config)#default-router 192.168.1.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#write memory
Building configuration...
[OK]
Router#

```

IP Configuration	
☒ DHCP	☒ Static
DHCP request failed.	
IPv4 Address	
Subnet Mask	
Default Gateway	0.0.0.0
DNS Server	0.0.0.0
IPv6 Configuration	

4. Simulate a Zomato-style restaurant WiFi scenario in Cisco Packet Tracer: configure a DHCP server to provide IP addresses to both a laptop and a smartphone (use a generic PC as smartphone). Show the assigned IPs and explain why DHCP is useful in this situation.

In a Zomato-style restaurant WiFi network, a **DHCP server** automatically provides IP addresses to devices such as laptops and smartphones. When a device connects to the WiFi, it sends a DHCP request and the server assigns an available IP address.

For example, the laptop may receive **192.168.10.10** and the smartphone may receive **192.168.10.11**.

Why DHCP is useful:

- It automatically assigns IP addresses to devices.
- It saves time because IP addresses do not need to be configured manually.
- It reduces IP address conflicts and configuration errors.
- It makes it easy to connect many customer devices to the restaurant WiFi.